



W4: Linear Model Assignment



Raj Kumar Biswokarma • Yesterday

Assignments • 100 points

Due 24 May

This Week's Tasks:

Statistical ML: Linear Models — Data Scientist at a Telecom Company

Objective: Go from raw customer data to a production-ready churn prediction recommendation. You will experiment with multiple linear models, evaluate honestly, and defend your final decision with evidence.

Key Tasks:

- Formulate the ML problem formally. Define X , y , loss function, and evaluation metric
- Train and compare Logistic Regression, Ridge Classifier, and SGD Classifier
- Predict customer tenure and compute Customer Lifetime Value (CLV)
- Run cross-validation, plot learning curves, and deliberately test for data leakage
- Write a model card for the model you would deploy

Requirements:

- There is no single right model — justify every choice with evidence from your experiments
- You are encouraged to use AI to debug and prototype — but you must be able to explain your code and decisions
- Log your failed experiments, surprising findings, and the moments where your intuition was wrong
- You decide which models to try, which metrics to report, and which model to ship

Dataset:

Telco Customer Churn <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>
Single CSV · 7,043 rows · 21 columns · Target: Churn (Yes/No)

TotalCharges has whitespace nulls · Class imbalance (~27% churn) · Mixed data types

See the attached assignment doc and student notebook below

Complete before our next session

W4_Linear_Model_A...
PDF



W4_Linear_Models_...
Unknown



Your work

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